THE CUMULATIVE NET HOURLY RATE MODEL

A Formal Derivation from First Principles

for Application to Private Hire Vehicle Operations

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Sentinel Project — Analytical Foundation

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§1 Motivation and Problem Statement

A Private Hire Vehicle (PHV) driver operating under a platform-mediated dispatch model (e.g., Uber) faces a fundamental economic question: at any given moment during the working week, what is the true rate of return on time invested? The answer is non-trivial because the driver's cost structure combines fixed weekly obligations with variable per-shift costs, both of which must be amortised across all hours worked before any net earnings are realised.

Conventional metrics reported by the platform (gross fare, trip duration, trip distance) are insufficient for this purpose. They exclude the fixed cost base entirely, they ignore unpaid time between trips (deadhead, idle waiting, repositioning), and they present each trip in isolation rather than as a contribution to a cumulative weekly position.

This document derives a model that answers the driver's question precisely. The model is constructed from first principles using only observable quantities, requires no forecasting or estimation, and is computable in real-time after each trip completion. It produces two outputs: a single scalar value representing the driver's current net hourly rate, and a time-series profile revealing the trajectory of that rate across the working week.

§1.1 The Central Question

Given a fixed weekly cost base, a sequence of trips with known earnings and durations, and a target net hourly rate, at what point in the week does the driver transition from sub-optimal to optimal earning, and what proportion of total hours worked falls into each category?

§1.2 Scope and Constraints

The model is deliberately minimal. It operates under four constraints:

Observability. Only quantities directly observable to the driver are used: trip earnings, trip duration (inclusive of all time consumed), and fixed costs known at the start of the week.

No forecasting. The model describes the driver's position at each point in time without predicting future trips, demand, or market conditions.

Real-time computability. The model updates after each trip with a single arithmetic operation. No iterative computation, no optimisation, no simulation.

Universality. The model applies to any driver with any cost structure on any platform. The specific parameters (rent, charging cost) are inputs, not assumptions.

§2 Axioms

The model rests on four axioms that formalise the economic structure of platform-mediated PHV driving.

Axiom 2.1 (Fixed Cost Precedence)**.** The driver incurs a fixed weekly cost C_w that must be fully recovered before any net earnings are realised. This cost exists independently of whether any trips are completed:

$$C_w \text{ is incurred at } t = 0 \text{ and is independent of } \{e_i, \tau_i\} \tag{1}$$

This axiom captures the reality of the rental model: the vehicle rental fee is due regardless of utilisation. It is the defining structural feature of the driver's economic position.

Axiom 2.2 (Time Scarcity)**.** The driver's sole productive resource is time. All economic activity consumes time, and consumed time cannot be recovered or reused within the same week:

$$T_n = \sum_{i=1}^{n} \tau_i \quad \text{is monotonically increasing and bounded by } |S| \tag{2}$$

where $|S|$ represents the total available shift time in the week.

Axiom 2.3 (Total Time Accounting)**.** The time cost of a trip includes *all* time consumed from the end of the previous trip (or start of shift) to the completion of the current trip:

$$\tau_i = \tau_i^{\text{dead}} + \tau_i^{\text{pickup}} + \tau_i^{\text{trip}} \tag{3}$$

This axiom rejects the platform's reported "trip time" as the relevant metric. The driver is on duty for the entirety of the interval, whether paid or unpaid.

Axiom 2.4 (Sequential Resolution)**.** Trips are completed in sequence. The driver's financial position can be evaluated after each trip completion, and the evaluation at trip n depends only on the cumulative totals through trip n:

$$r_n = f(E_n, T_n, C_w) \quad \text{where } E_n = \sum_{i=1}^{n} e_i, \quad T_n = \sum_{i=1}^{n} \tau_i \tag{4}$$

This axiom permits real-time computation: no future information is required.

§3 Definitions

§3.1 Cost Structure

Definition 3.1 (Vehicle Rental)**.** The weekly rental cost C_{rent} is a fixed obligation payable regardless of utilisation. For the current operating context: $C_{\text{rent}} = £220$.
Definition 3.2 (Charging Cost)**.** The daily charging cost C_{charge} is the cost of replenishing the vehicle's energy supply, treated as a fixed daily cost using the upper estimate to avoid understatement. $C_{\text{charge}} = £30/\text{day}$.
Definition 3.3 (Weekly Fixed Cost)**.** The total weekly cost is the sum of rental and charging obligations:

$$C_w = C_{\text{rent}} + 7 \times C_{\text{charge}} = 220 + 7(30) = £430 \tag{5}$$

This formulation assumes 7 days of charging at the upper bound. The critical design choice is to front-load all costs at the start of the week, establishing the full deficit the driver must recover.

§3.2 Trip Observables

Definition 3.4 (Trip Earnings)**.** For trip i, the total earnings e_i represent the complete payment received by the driver, inclusive of base fare, surge multiplier, tips, holiday pay, and any per-trip bonuses.
Definition 3.5 (Trip Time)**.** For trip i, the total time τ_i is the complete duration in hours from the moment the driver becomes available for that trip to the moment the current trip is completed:

$$\tau_i = \frac{\tau_i^{\text{dead}} + \tau_i^{\text{pickup}} + \tau_i^{\text{trip}}}{60} \quad \text{[converting minutes to hours]} \tag{6}$$

§3.3 Cumulative Quantities

Definition 3.6 (Cumulative Earnings)**.** The total earnings after n trips:

$$E_n = \sum_{i=1}^{n} e_i \tag{7}$$

Definition 3.7 (Cumulative Time)**.** The total time consumed after n trips:

$$T_n = \sum_{i=1}^{n} \tau_i \tag{8}$$

Both quantities are monotonically increasing: $E_n \geq E_{n-1}$ (earnings cannot be negative) and $T_n > T_{n-1}$ (every trip consumes positive time).

§4 Derivation of the Core Model

§4.1 The Net Position

At any point during the week, after n trips, the driver's net financial position is:

$$\text{Net}_n = E_n - C_w \tag{9}$$

This quantity is negative at the start of the week (£430 deficit) and increases with each trip. The sign determines the driver's phase.

§4.2 The Cumulative Net Hourly Rate

Theorem 4.1 (Core Equation)**.** *The net hourly rate at trip n is the net position divided by total time worked:*

$$\boxed{r_n = \frac{E_n - C_w}{T_n} = \frac{\sum_{i=1}^{n} e_i - 430}{\sum_{i=1}^{n} \tau_i}} \tag{10}$$

This is the core equation of the model. It has one parameter ($C_w = 430$), two running totals (E_n and T_n), and no unknowns. It is computable after each trip with $O(1)$ operations.

§4.3 Initial Condition

At $n = 0$ (start of week, no trips completed):

$$r_0 = \frac{0 - 430}{0} = \text{undefined} \tag{11}$$

The expression is undefined due to division by zero. Conceptually, this represents an infinitely negative rate: the driver owes £430 with zero hours invested. The model is initialised at $n = 1$.

§4.4 Recursive Formulation

Theorem 4.2 (Recursive Update)**.** *Given r_n after trip n, the rate after trip $n + 1$ is:*

$$r_{n+1} = \frac{E_n + e_{n+1} - C_w}{T_n + \tau_{n+1}} \tag{12}$$

Remark. Each new trip pulls the running rate toward the instantaneous rate of that trip (e_{n+1}/τ_{n+1}), with pull strength inversely proportional to accumulated time T_n. Early trips dominate the trajectory; later trips have diminishing marginal influence.

§4.5 The Instantaneous Trip Rate

For reference, the instantaneous rate of trip i in isolation is:

$$\rho_i = \frac{e_i}{\tau_i} \tag{13}$$

This is what the platform implicitly reports. It bears no relationship to the driver's true net position because it excludes the cost base entirely. The divergence between ρ_i and r_n is the entire motivation for this model.

§5 Properties of the Model

§5.1 Phase Structure

The sign of r_n partitions the working week into three phases:

Phase	Condition	Interpretation
Deficit	$E_n < C_w \;\Rightarrow\; r_n < 0$	Driver has not recovered fixed costs
Break-even	$E_n = C_w \;\Rightarrow\; r_n = 0$	Costs exactly recovered
Profit	$E_n > C_w \;\Rightarrow\; r_n > 0$	Every subsequent hour generates true income

Every driver's week begins in the Deficit phase and, if sufficient trips are completed, transitions through Break-even into Profit.

§5.2 Break-Even Point

Definition 5.1 (Break-Even Trip)**.** The break-even trip n^* is the first trip at which cumulative earnings equal or exceed the weekly cost:

$$n^* = \min\{n : E_n \geq C_w\} \tag{14}$$

The corresponding break-even time $T^* = T_{n^*}$ represents the hours worked before net earning begins. This is the single most important diagnostic metric.

§5.3 Asymptotic Behaviour

Theorem 5.1 (Convergence)**.** *If trips are drawn from a stationary distribution with mean earnings $\bar{e}$ and mean duration $\bar{\tau}$, then:*

$$\lim_{n\to\infty} r_n = \frac{\bar{e}}{\bar{\tau}} - \frac{C_w}{T_n} \;\longrightarrow\; \frac{\bar{e}}{\bar{\tau}} = \rho \tag{15}$$

The net hourly rate converges to the gross instantaneous trip rate ρ. However, convergence is hyperbolic ($1/T$), not exponential, meaning the cost base depresses the net rate for a substantial portion of the working week.

§5.4 Marginal Impact Decay

Proposition 5.1 (Sensitivity)**.** *The partial derivatives of r_n with respect to individual trip variables are:*

$$\frac{\partial r_n}{\partial e_i} = \frac{1}{T_n} \tag{16}$$

$$\frac{\partial r_n}{\partial \tau_i} = -\frac{E_n - C_w}{T_n^2} \tag{17}$$

The sensitivity to any single trip's earnings decreases as $1/T_n$. Early trips, when T_n is small, have outsized influence. This has a critical operational implication: the first few trips of the week disproportionately determine the week's financial outcome.

§5.5 The Idealised Continuous Approximation

Assuming a constant gross trip rate $\rho = \bar{e}/\bar{\tau}$, the discrete model admits a continuous approximation:

$$r(T) = \rho - \frac{C_w}{T} \tag{18}$$

This is a rectangular hyperbola displaced vertically by ρ, with vertical asymptote at $T = 0$ and horizontal asymptote at $r = \rho$. All empirical rate profiles are perturbations of this curve. The continuous form is used for analytical results; the discrete form for empirical computation.

§6 The Target Rate and Optimal/Sub-Optimal Classification

§6.1 The Target Net Rate

Definition 6.1 (Target Rate)**.** The target net hourly rate r^* is the minimum acceptable cumulative net rate the driver considers satisfactory. It is a driver-defined parameter reflecting personal financial requirements, opportunity cost of time, and quality-of-life considerations.

The target rate is not derived from the model; it is an exogenous input. The model's role is to determine when the target is achieved and what proportion of hours fall above or below it.

§6.2 The Threshold Hour

Under the continuous approximation, the threshold hour T^* is the time at which the cumulative net rate first reaches the target:

$$\begin{aligned} r(T^*) = r^* \implies & \rho - \frac{C_w}{T^*} = r^* \\ \implies & \boxed{T^* = \frac{C_w}{\rho - r^*}} \end{aligned} \tag{19}$$

Three immediate implications follow:

1. T^* is **directly proportional** to C_w: doubling costs doubles hours to target.
2. T^* is **inversely proportional** to the margin $(\rho - r^*)$: the closer the target is to the gross rate, the longer it takes.
3. T^* is **undefined** when $\rho \leq r^*$: if gross rate does not exceed target, the target is mathematically unachievable.

§6.3 Classification of Hours

Definition 6.2 (Sub-Optimal Hour)**.** An hour T is sub-optimal if $r(T) < r^*$, equivalently $T < T^*$.
Definition 6.3 (Optimal Hour)**.** An hour T is optimal if $r(T) \geq r^*$, equivalently $T \geq T^*$.

Every hour before T^* is sub-optimal; every hour from T^* onward is optimal. This is a binary classification with a single transition point.

§6.4 The Proportion

For a total weekly working time T_w, the proportions are:

$$P_{\text{sub}} = \frac{T^*}{T_w} = \frac{C_w}{T_w(\rho - r^*)} \tag{20}$$

$$P_{\text{opt}} = 1 - P_{\text{sub}} = 1 - \frac{C_w}{T_w(\rho - r^*)} \tag{21}$$

These proportions are the model's primary diagnostic output. They quantify the structural efficiency of the driver's week: what fraction of time is spent servicing costs versus earning at an acceptable rate.

§6.5 Baseline Calculation

Using current operating parameters (C_w = £430, ρ = £25/hr, r^* = £15/hr):

$$T^* = \frac{430}{25 - 15} = 43 \text{ hours} \tag{22}$$

Weekly Hours T_w	P_{sub}	P_{opt}	**Sub-Optimal Hrs**	**Optimal Hrs**
50	86.0%	14.0%	43.0	7.0
55	78.2%	21.8%	43.0	12.0
60	71.7%	28.3%	43.0	17.0
65	66.2%	33.8%	43.0	22.0
70	61.4%	38.6%	43.0	27.0
80	53.8%	46.3%	43.0	37.0
86	50.0%	50.0%	43.0	43.0

The driver must work **86 hours per week** to achieve parity between sub-optimal and optimal hours at these parameters.

§7 Sensitivity to Cost Reduction

§7.1 The Cost Lever

The threshold hour is linearly proportional to C_w:

$$\frac{\partial T^*}{\partial C_w} = \frac{1}{\rho - r^*} \tag{23}$$

At $\rho = £25$ and $r^* = £15$: every £10 reduction in weekly cost reduces T^* by 1 hour.

§7.2 Cost Reduction Scenarios

Scenario	C_w	T^*	P_{sub} **at 60h**	P_{opt} **at 60h**
Current (rental + charging)	£430	43.0 hrs	71.7%	28.3%
Reduce charging to £20/day	£360	36.0 hrs	60.0%	40.0%
Own vehicle, £30/day charge	£210	21.0 hrs	35.0%	65.0%
Own vehicle, £20/day charge	£140	14.0 hrs	23.3%	76.7%
Own vehicle, free charging	£0	0.0 hrs	0.0%	100.0%

Vehicle ownership is the single largest available intervention. Eliminating the £220/week rental immediately shifts T^* leftward by 22 hours.

§7.3 The Cost–Hour Equivalence

Theorem 7.1 (Cost–Rate Exchange)**.** *A reduction of ΔC in weekly costs is equivalent to a reduction of $\Delta C/(\rho - r^*)$ hours in sub-optimal time. The equivalent gross rate improvement is:*

$$\Delta\rho_{equiv} = \frac{\Delta C \cdot (\rho - r^*)}{C_w - \Delta C} \tag{24}$$

At current values, a £50/week rental reduction is equivalent to earning an additional £1.32/hr gross on every trip.

§7.4 The Parity Condition

Corollary 7.1 (Equal Split)**.** *For $P_{opt} = P_{sub} = 50\%$:*

$$T_w = \frac{2C_w}{\rho - r^*} \tag{25}$$

*At current parameters: $T_w = 86$ hours. With vehicle ownership ($C_w = 210$): $T_w = 42$ hours. The structural difference is **44 hours per week** — the mathematical cost of the rental model.*

§8 Expected Rate Profiles

§8.1 The General Shape

Every weekly profile follows the same fundamental arc: beginning at a large negative value, climbing through zero at break-even, and asymptotically approaching ρ. Under the continuous approximation $r(T) = \rho - C_w/T$, this is a rectangular hyperbola. All real profiles are perturbations of this curve driven by trip-to-trip variability.

§8.2 Profile Taxonomy

Profile A — Strong Start. Lucrative early trips produce rapid ascent from deficit. Break-even occurs early, the curve is steep and smooth, and the rate stabilises quickly. This is the optimal profile: fixed costs are amortised rapidly, and subsequent trips contribute almost entirely to profit.

Profile B — Steady Grind. Uniform trip quality produces a profile that closely follows the idealised hyperbola. Break-even is predictable. Sustainable but offers limited margin for disruption.

Profile C — Slow Start, Late Recovery. Poor early trips leave the profile deeply negative for an extended period. A late surge produces a sharp upward inflection. Operationally fragile: depends on late-week performance to salvage the position.

Profile D — Plateau Stagnation. The driver works consistently but at a low instantaneous rate. The profile flattens prematurely, approaching an asymptote below target. This indicates a structural problem: ρ is insufficient. More hours cannot resolve this — only improved trip quality can.

Profile E — Volatile. Pronounced sawtooth behaviour. The overall trend is upward but erratic. The variance quantifies exposure to trip quality risk.

Profile F — Deteriorating. The profile climbs initially but then declines. This occurs when $e_{n+1}/\tau_{n+1} < r_n$: each additional trip *reduces* the cumulative rate. An alarm signal indicating the driver should cease working.

§8.3 Diagnostic Indicators

Indicator	What It Reveals	Action
Break-even time T^*	Hours until positive net earning	Earlier is better
Gradient at break-even	Momentum entering profit phase	Steep = strong position
Convergence rate	How quickly rate stabilises	Fast = reliable rhythm
Downward inflection	Current trips below running average	Consider ending shift

§9 Optimal Hours Determination

The rate function $r(T)$ is always increasing, so a simple maximum does not exist. The optimum must be defined by alternative statistical criteria.

§9.1 The Marginal Threshold Method

Define a minimum acceptable marginal improvement ε (£/hr per additional hour worked). Stop when:

$$r'(T_\varepsilon^*) = \frac{C_w}{(T_\varepsilon^*)^2} = \varepsilon \quad \Longrightarrow \quad \boxed{T_\varepsilon^* = \sqrt{\frac{C_w}{\varepsilon}}} \tag{26}$$

Threshold ε	**Interpretation**	**Optimal Hours**
£1.00/hr per hr	Aggressive — stop early	20.7
£0.50/hr per hr	Moderate	29.3
£0.25/hr per hr	Patient	41.5
£0.10/hr per hr	Very patient	65.6

§9.2 The Proportion-of-Asymptote Method

Set a target efficiency ratio η^* (fraction of maximum achievable rate):

$$\eta(T) = \frac{r(T)}{\rho} = \eta^* \quad \Longrightarrow \quad \boxed{T_\eta^* = \frac{C_w}{\rho(1-\eta^*)}} \tag{27}$$

Target η^*	**T^* at $\rho = £20$**	**T^* at $\rho = £25$**	**T^* at $\rho = £30$**
60%	53.8 hrs	43.0 hrs	35.8 hrs
70%	71.7 hrs	57.3 hrs	47.8 hrs
75%	86.0 hrs	68.8 hrs	57.3 hrs
80%	107.5 hrs	86.0 hrs	71.7 hrs

§9.3 The Knee Point Method

The point of maximum curvature on $r(T)$. For the idealised case:

$$T_{\text{knee}} \approx \sqrt{C_w} \approx 20.7 \text{ hours} \tag{28}$$

This coincides approximately with break-even and is typically too aggressive for practical use.

§9.4 Practical Recommendation

The marginal threshold method is most operationally meaningful. The recommended Sentinel formulation is:

$$\text{Continue working} \iff \frac{C_w}{T_{\text{current}}^2} > \varepsilon_{\text{driver}} \tag{29}$$

where $\varepsilon_{\text{driver}}$ is a personal preference parameter.

§10 Empirical Application

§10.1 Data Requirements

To compute the empirical rate profile and transition point for any given week:

Field	Description	Source
Week identifier	ISO week or date range	Derived from trip date
Trip completion time	Clock time of trip end	Uber trip data (JSON)
Trip earnings e_i	Total payment (£)	Uber trip data
Total trip time τ_i	Dead + pickup + trip (hours)	Uber trip data + GPS

§10.2 Computation Procedure

For each week:

1. Order all trips chronologically.
2. Initialise $E_0 = 0$, $T_0 = 0$.
3. For each trip $i = 1, 2, \ldots, n$:
   (a) Update $E_i = E_{i-1} + e_i$.
   (b) Update $T_i = T_{i-1} + \tau_i$.
   (c) Compute $r_i = (E_i - C_w)/T_i$.
   (d) Record $(i,\ t_{\text{clock},i},\ E_i,\ T_i,\ r_i)$.
4. Identify $n^* = \min\{i : r_i \geq r^*\}$ and record the transition clock time and T^*.
5. Compute $P_{\text{sub}} = T^*/T_w$ and $P_{\text{opt}} = 1 - P_{\text{sub}}$.

§10.3 Cross-Week Comparison

With multiple weeks computed, the model produces a distribution of transition times. The mean, variance, and range of T^* across weeks characterise the driver's structural position: a low mean with low variance indicates consistently efficient weeks; a high mean with high variance indicates erratic performance dominated by the cost structure.

§10.4 Pending: Data Application

The December 2025 – January 2026 trip data (previously analysed in JSON format) will be applied to this model to produce empirical rate profiles, transition points, and $P_{\text{sub}}/P_{\text{opt}}$ distributions for each observed week. This analysis is pending re-upload of the trip data files.

§11 Structural Implications

§11.1 The Rental Extraction

The model makes explicit what is often felt intuitively: the rental model structurally condemns the driver to spending the majority of working hours servicing costs rather than earning. P_{sub} is not a reflection of driver skill or effort — it is a mathematical consequence of the cost structure. At current parameters, a 60-hour week yields only 17 optimal hours (28.3%). The driver works 43 hours before earning a single pound at target rate.

§11.2 Three Levers

The model identifies three and only three levers for improving the distribution:

Reduce C_w. Most direct. Vehicle ownership eliminates £220/week, reducing T^* by 22 hours. Cheaper charging reduces T^* by £10/$(\rho - r^*)$ hours per £10 saved. This is the highest-impact intervention.

Increase ρ. Improve gross trip rate through better trip selection, surge capture, and reduced deadhead. Each £1/hr improvement reduces T^* by $C_w/(\rho - r^*)^2$ hours. This is the domain of Sentinel's decision engine.

Lower r^*. Accept a lower target rate. This is a concession, not an optimisation. It shifts the threshold leftward by reducing the denominator $(\rho - r^*)$, but does so by lowering expectations rather than improving economics.

§11.3 Integration with Sentinel

This model provides Sentinel with a real-time diagnostic of shift and week quality requiring no predictive model. After each trip, Sentinel reports r_n, the distance from target $(r_n - r^*)$, and the profile shape. Combined with the trip acceptance model and surge positioning strategy derived separately, it completes the analytical foundation for intelligent shift management.

End of Document

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